MODULE 30 · DOMAIN 5 - NETWORKING AND CONTENT DELIVERY

Private and Hybrid Connectivity

VPC endpoints, PrivateLink, peering, Transit Gateway, Site-to-Site VPN and Direct Connect.

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Four questions: reach an **AWS service** privately, join **two** VPCs, join **many**, or reach **on-premises**. Each has one obvious answer.

The options, mapped

REQUIREMENT	ANSWER
Reach S3 or DynamoDB privately from a VPC	Gateway endpoint. Free
Reach another AWS service privately	Interface endpoint, powered by PrivateLink
Expose your own service privately to other accounts	PrivateLink endpoint service, behind an NLB
Connect two VPCs	VPC peering
Connect many VPCs and on-premises networks	Transit Gateway
Reach a data centre over the internet	Site-to-Site VPN
Reach a data centre over a private circuit	Direct Connect

ON THE EXAM

Gateway endpoints exist for exactly two services. Everything else private within a Region is an interface endpoint, and that is charged.

Gateway or interface endpoint?

GATEWAY ENDPOINT

S3 and DynamoDB only

- A route table entry pointing at a prefix list
- Completely free, with no data processing charge
- Same Region only; no ENI and no security group
- NOT reachable from on-premises over DX or VPN

INTERFACE ENDPOINT

Almost every other service

- An ENI with a private IP in your subnet
- Charged per AZ per hour, plus per GB processed
- Has a security group, so you control access to it
- Reachable from on-premises over DX or VPN

THE TIEBREAKER

The last row of each column is the deciding one. If on-premises systems must reach S3 privately, a gateway endpoint cannot do it; an interface endpoint can.

Endpoint policies and private DNS

- An **endpoint policy** restricts what may be done through that endpoint: only these buckets, only these actions, only this organisation.
- That is a **data perimeter** control. Traffic to a bucket outside your organisation can be blocked at the endpoint.
- **Private DNS** on an interface endpoint makes the service's normal public hostname resolve to the endpoint's private address.
- Private DNS needs **enableDnsSupport and enableDnsHostnames** on the VPC. Module 28 flagged those two attributes.

> ON THE EXAM
>
> Private DNS is what makes an interface endpoint transparent. Without it, applications must be changed to use the endpoint-specific hostname.

PrivateLink for your own services

- Put your service behind a **Network Load Balancer** or Gateway Load Balancer, then create an **endpoint service**.
- Consumers create an interface endpoint to it. Traffic never traverses the internet, and **no peering or route sharing is required**.
- **CIDR ranges may overlap**, because there is no routing between the two VPCs. That is the significant advantage over peering.
- The provider controls **allowed principals** and can require manual acceptance of each connection request.

> ON THE EXAM
>
> PrivateLink is unidirectional: the consumer reaches the provider, never the reverse. That asymmetry is exactly what makes it safe for SaaS.

VPC peering

- A one-to-one connection between two VPCs. Works **across accounts and across Regions**, using the AWS backbone.
- **CIDR ranges must not overlap.** This is the constraint that kills peering at scale, and module 28's IPAM prevents it.
- **Peering is not transitive.** If A peers with B and B peers with C, A cannot reach C.
- **No edge-to-edge routing.** You cannot use a peer's internet gateway, NAT gateway, VPN or Direct Connect connection.

> ON THE EXAM
>
> Both sides need routes. A peering connection that is Active with traffic still failing is almost always a route table nobody updated on one side.

Transit Gateway

- A regional router. Everything **attaches once** to it: VPCs, VPN connections, Direct Connect gateways, and other Transit Gateways.
- Replaces a mesh. Ten VPCs need **45 peering connections** or **10 attachments**.
- **Route tables** on the gateway control who can reach whom. **Association** decides which table an attachment uses; **propagation** decides which routes it advertises.
- That separation is how you build isolation: a shared services VPC reachable by everyone, while spokes cannot reach each other.

> ON THE EXAM
>
> Association and propagation are different things and the exam tests both. One is which routes you see; the other is which routes you publish.

Transit Gateway details that decide questions

- **Appliance mode** keeps both directions of a flow on the same Availability Zone, which stateful firewalls require.
- Without it, asymmetric routing breaks inspection appliances, and the symptom is intermittent connection failures.
- **Inter-Region peering** connects Transit Gateways in different Regions, encrypted, over the AWS backbone.
- **Security group referencing support** must be enabled on an attachment, as module 29 covered, or group references are discarded.

> ON THE EXAM
>
> Appliance mode is the answer whenever a stateful inspection appliance sits behind a Transit Gateway. Without it, the return traffic can arrive at a different appliance.

Peering or Transit Gateway?

PEERING WHEN

A small, fixed number

- Two or three VPCs with a stable relationship
- No data processing charge, only data transfer
- Lowest latency: no intermediate hop
- Becomes unmanageable beyond a handful

TRANSIT GATEWAY WHEN

Many, or hybrid

- Dozens of VPCs, or VPN and Direct Connect too
- Central route tables give real segmentation
- Charged per attachment per hour, plus per GB
- One extra hop of latency

THE TIEBREAKER

Peering has no data processing charge, so a very high volume link between two VPCs can be cheaper peered even in an estate that otherwise uses a Transit Gateway.

Site-to-Site VPN

- IPsec over the internet. Every connection has **two tunnels** to two AWS endpoints, for redundancy.
- **Your device must be configured for both.** A single-tunnel configuration loses connectivity during AWS tunnel maintenance.
- Terminates on a **virtual private gateway** or a **Transit Gateway**. The customer gateway represents your device.
- **BGP** gives dynamic routing and automatic failover. Static routing works and fails over only when you intervene.

> ON THE EXAM
>
> Two tunnels are provided and only one is used unless you configure both. That is the most common cause of an unexpected VPN outage.

New: VPN bandwidth changed twice

- **November 2025: large bandwidth tunnels** support **5 Gbps per tunnel**, four times the standard 1.25 Gbps.
- Available on **Transit Gateway or Cloud WAN only**, not on a virtual private gateway.
- It **reduces the need for ECMP**, which was previously the only way to aggregate bandwidth across bonded tunnels.
- **May 2026:** bandwidth can be **changed on an existing connection**, preserving tunnel IPs, CIDRs and pre-shared keys. Previously this meant delete and recreate.

> ON THE EXAM
>
> The May 2026 change removes the reason teams avoided upgrading: no new tunnel IPs, no on-premises device reconfiguration, no firewall changes.

Other VPN options

- **VPN Concentrator**, from November 2025, connects up to **100 low-bandwidth remote sites** through a **single Transit Gateway attachment**.
- For a retailer with hundreds of branches, that replaces hundreds of individual attachments.
- **Accelerated VPN** routes tunnel traffic through **AWS Global Accelerator**, entering the AWS network at the nearest edge location.
- **ECMP** on a Transit Gateway still aggregates bandwidth across multiple tunnels, up to around 50 Gbps.

> ON THE EXAM
>
> Accelerated VPN shortens the distance travelled over the public internet, which is where jitter and packet loss come from on a long-haul VPN.

Direct Connect

- A private physical circuit into an AWS Direct Connect location. **Dedicated** connections at 1, 10, 100 or 400 Gbps, ordered from AWS.
- **Hosted** connections from a partner run from 50 Mbps to 25 Gbps, and are provisioned far faster.
- Three **virtual interfaces**: **public** for AWS service endpoints, **private** for a VPC via a VGW or DX gateway, and **transit** for a Transit Gateway.
- **Direct Connect is not encrypted by default.** Add **MACsec** on supported 10 and 100 Gbps dedicated connections, or run a VPN over it.

> ON THE EXAM
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> Private is not the same as encrypted. That distinction appears in compliance questions and the answer is MACsec or an IPsec VPN on top.

Direct Connect at scale

- A **Direct Connect gateway** is global, letting one connection reach VPCs in **multiple Regions**. Up to **20 VGWs or 6 Transit Gateways**.
- A **link aggregation group** bundles connections into one logical link: up to 4 below 100 Gbps, or 2 at 100 Gbps.
- **SiteLink** sends traffic directly between two Direct Connect locations over the AWS backbone, **bypassing Regions entirely**.
- Resiliency: **High Resiliency** is two connections at two locations. **Maximum Resiliency** is four across two locations.

> ON THE EXAM
>
> SiteLink turns Direct Connect into a private global WAN between your own sites, which is a use case that has nothing to do with reaching AWS.

Direct Connect or VPN?

DIRECT CONNECT WHEN

Consistent and high volume

- Predictable latency and dedicated bandwidth
- Lower data egress pricing
- Weeks to provision, and a fixed port cost
- Not encrypted unless you add MACsec or IPsec

VPN WHEN

Fast, encrypted, flexible

- Live in under an hour, over any internet connection
- Encrypted by default with IPsec
- Now up to 5 Gbps per tunnel on a Transit Gateway
- Subject to internet variability

THE TIEBREAKER

The usual production answer is both: Direct Connect as primary and a VPN as automatic backup. When both advertise the same routes, Direct Connect is always preferred.

> On-premises servers connected over Direct Connect must reach an S3 bucket privately, without traversing the internet. The team creates a gateway endpoint for S3 in the VPC and adds it to the route tables, but on-premises traffic still cannot reach S3 privately.
>
> In-VPC instances reach S3 through the endpoint correctly.

What should the team do?

A. Create an interface endpoint for S3, which can be reached from on-premises over Direct Connect

B. Add the on-premises CIDR range to the gateway endpoint's policy

C. Propagate the gateway endpoint's prefix list over BGP

D. Create a second gateway endpoint in the Direct Connect gateway

◆ PAUSE AND DECIDE

What should the team do?

A Interface endpoint for S3

Gateway endpoints cannot be reached from on-premises over Direct Connect or VPN. Interface endpoints have a private IP in your subnet, and can.

B Endpoint policy change

An endpoint policy restricts what may be done through an endpoint that is already reachable. It cannot make an unreachable one reachable.

C Propagate the prefix list

Gateway endpoints work through a route table entry inside the VPC. There is no path to advertise to an on-premises router.

D Gateway endpoint on the DX gateway

Gateway endpoints attach to a VPC route table. There is no such construct on a Direct Connect gateway.

THE SIGNAL

The controlled comparison is stated: in-VPC access works, on-premises does not. That is precisely the documented boundary of a gateway endpoint.

> A software vendor must expose an API to 200 customer AWS accounts privately. Many customers use the 10.0.0.0/16 range, which overlaps with the vendor's own VPC. Customers must not be able to reach anything else in the vendor's environment.
>
> The vendor does not want to manage 200 peering connections.

What should the vendor implement?

A. A PrivateLink endpoint service behind a Network Load Balancer, with customers creating interface endpoints

B. VPC peering with each customer account

C. A Transit Gateway shared with customers through RAM

D. A public API endpoint restricted by IP allow-list

◆ PAUSE AND DECIDE

What should the vendor implement?

A PrivateLink endpoint service

No routing between VPCs, so overlapping CIDRs are irrelevant. Exposure is limited to the one service, and it scales to any number of consumers.

B Peering per customer

Fails on overlapping CIDRs alone, and 200 peering connections is exactly the management burden the vendor ruled out.

C Shared Transit Gateway

Creates routing between the vendor and customers, so overlaps break it and customers gain a route into the vendor's network.

D Public endpoint with allow-list

Traffic traverses the internet, which the requirement excludes, and an IP allow-list for 200 customers is unmaintainable.

THE SIGNAL

Overlapping CIDRs is the clause that eliminates everything routing-based. PrivateLink works precisely because it does not route.

> Traffic between two VPCs attached to a Transit Gateway passes through stateful firewall appliances. Those sit in an inspection VPC, spread across three Availability Zones. Connections succeed most of the time and fail intermittently, with no pattern the team can identify.
>
> The appliances log packets arriving without a matching session.

What is the cause?

A Appliance mode is not enabled on the inspection VPC's Transit Gateway attachment

B The Transit Gateway route tables are missing a propagation

C Security group referencing support is not enabled

D The appliances need larger instance types

◆ PAUSE AND DECIDE

What is the cause?

A Appliance mode not enabled

Without it the Transit Gateway may send the return path through a different AZ, so a stateful appliance sees a packet for a session it never opened.

B Missing propagation

Would break connectivity consistently, not intermittently. The scenario says most connections succeed.

C SG referencing support

A real Transit Gateway setting, and it would cause consistent denial for group-referencing rules, not asymmetric flows.

D Bigger appliances

The log message names a missing session, not exhausted resources. Capacity would show as drops under load, not orphaned packets.

THE SIGNAL

"Packets arriving without a matching session" is the asymmetric-routing signature. Intermittent rather than consistent points at path selection, not configuration.

> A company runs a Site-to-Site VPN to a Transit Gateway. Two tunnels use ECMP, achieving about 2.5 Gbps. A migration now needs 5 Gbps sustained. The team is reluctant to change anything, because the last bandwidth change required deleting and recreating the connection.
>
> It is 2026.

What should the team do?

A. Modify the existing connection's tunnel bandwidth to large, which preserves tunnel IPs and pre-shared keys

B. Add two more tunnels and extend ECMP

C. Delete and recreate the connection with large bandwidth tunnels

D. Order a Direct Connect circuit

◆ PAUSE AND DECIDE

What should the team do?

A Modify bandwidth in place

Since May 2026 bandwidth can be changed on an existing connection, preserving tunnel IPs, CIDRs and pre-shared keys. No on-premises change at all.

B More tunnels with ECMP

Works, and it is the workaround that large bandwidth tunnels were introduced to remove. More tunnels means more device configuration to maintain.

C Delete and recreate

The team's stated fear, and it was the only option until May 2026. It is now unnecessary.

D Direct Connect

Takes weeks to provision and a migration is already under way. It is the right long-term answer to a different question.

THE SIGNAL

The team's reluctance is based on a constraint that was removed in May 2026. The phrase "it is 2026" is there to make the currency the point.

> A company has 14 VPCs across three accounts, currently joined by 21 peering connections. Adding a VPC now requires updating route tables in every existing VPC. They also need to connect two on-premises data centres, and want central control over which VPCs can reach each other.
>
> Address ranges do not overlap.

What should the company implement?

A. A Transit Gateway, shared across accounts with RAM, with route tables providing segmentation

B. Continue with peering and automate route table updates

C. PrivateLink endpoint services between each pair of VPCs

D. One large shared VPC replacing all 14

◆ PAUSE AND DECIDE

What should the company implement?

A Transit Gateway shared via RAM

One attachment per VPC, VPN attachments for the data centres, and route tables giving the central segmentation asked for. All three requirements.

B Automate peering routes

Automates the symptom. Peering is non-transitive and cannot attach a VPN, so the on-premises requirement is unmet regardless.

C PrivateLink between pairs

Exposes individual services, not networks. It cannot provide general VPC-to-VPC routing or reach a data centre.

D One shared VPC

A genuine pattern, and it discards the isolation boundaries entirely and is an enormous migration for 14 existing VPCs.

THE SIGNAL

Three requirements, and peering fails two of them structurally: it cannot attach a VPN, and it offers no central place to control who reaches whom.

The five things to carry forward

- **Gateway endpoints cannot be reached from on-premises.** Interface endpoints can, and are charged.
- **PrivateLink does not route**, so overlapping CIDRs do not matter. Peering and Transit Gateway both do.
- **Peering is non-transitive and cannot attach a VPN.** Those are structural limits, not scale ones.
- **Appliance mode** pins both directions of a flow to one AZ. Intermittent failures with stateful appliances mean it is missing.
- **VPN tunnels now reach 5 Gbps**, and since May 2026 bandwidth changes in place with no on-premises reconfiguration.

> ON THE EXAM
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> Next: Route 53 - hosted zones, alias records, routing policies, Resolver and DNS Firewall.